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A multi-modal brain image fusion technique using nakagami imaging and intuitionistic fuzzy sets

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HIGHLIGHTS

- NI effectively reveals subtle brain lesions, even in low-contrast areas.
- NI creates three channels, enhancing each separately to increase the overall contrast.
- Innovating IFG aims to enhance details in NI channels for fusion.

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ABSTRACT

Medical image fusion is essential for consolidating images from diverse modalities into a single image, offering comprehensive information for diagnosis and analysis. With the rapid evolution of brain imaging technologies, researchers increasingly focus on refining fuzzy fusion techniques to utilize the full potential of these modalities. However, existing fuzzy fusion approaches often face three key challenges: difficulty in enhancing low-contrast images, uncertainty in selecting appropriate fuzzy membership functions, and redundancy in preserving important features. To address these shortcomings, a Nakagami distribution and intuitionistic fuzzy set-based multi-modal brain image fusion (NIFMBF) framework is proposed. This method integrates three core innovations: First, Nakagami imaging (NI) is employed to enhance low-contrast areas and reveal imperceptible lesions; second, a novel intuitionistic fuzzy generator (IFG) is designed to transform NI outputs into intuitionistic fuzzy images that effectively handle vagueness and enhance fine structural details; third, gray-level co-occurrence matrix (GLCM)-based contrast feature extraction guides the fusion process, enabling the retention of crucial information while reducing redundant data. Extensive experiments conducted on three benchmark datasets and compared against five state-of-the-art fusion techniques demonstrate the superior performance of the proposed NIFMBF method. Quantitatively, the method achieves an average improvement of 12–20% for every considered fusion metric. Additionally, the proposed framework reduces computational time by nearly 30% highlighting its efficiency. These findings, validated by both qualitative and quantitative evaluations, underscore the efficacy and potential of NIFMBF for medical image fusion.

1. Introduction

In recent years, there has been a rapid advancement in brain imaging technologies, offering various modalities customized to specific imaging needs. For instance, computed tomography (CT) provides a detailed view of rigid brain structures, while magnetic resonance imaging (MRI) reveals soft tissue structures. MRI offers a range of imaging modalities, such as T1-weighted MRI, T2-weighted MRI, PD-MRI, and GAD-MRI, with its ability to alter radio frequency signals, each providing unique insights into brain anatomy and pathology [1]. While each modality offers distinct advantages, they also come with limitations, and images may be susceptible to distortion due to environmental factors during acquisition. Brain image fusion is essential for early diagnosis because it combines the strengths of multiple modalities into a single informative image. It also reduces storage requirements [2]. Over time, numerous image fusion algorithms have been developed at pixel, feature, and decision levels. Further, pixel-level fusion techniques can be broadly divided into spatial domain and transform domain approaches. The spatial domain methods, such as arithmetic operations, minimum, maximum, median, and rank operators are computationally simple but

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often suppress important details and introduce noise [3]. Furthermore, statistical subspace methods in the spatial domain approach, including principal component analysis (PCA) [4], independent component analysis (ICA), and canonical correlation analysis (CCA) [5], improve feature extraction but may cause spectral degradation. On the other hand, transform domain methods have gained wider attention due to their ability to capture multi-resolution features. Early examples include Gaussian [6] and Laplacian [7] pyramid decompositions, but these approaches sacrifice spatial details. Wavelet-based methods, including discrete wavelet transform (DWT) [8], stationary wavelet transform (SWT) [9], and dual-tree complex wavelet transform (DTCWT), struggle with a lack of shift invariance and poor directional selectivity. Later multi-scale transforms like curvelet [10], contourlet [11], and shearlet [12] better preserved edges but still degraded gray values. To overcome these drawbacks, hybrid approaches combining spatial and transform methods have been explored [13,14]. Beyond these, strategic fusion methods such as guided filtering [15], fuzzy logic [16], dictionary learning, sparse representation [17], and neural networks [18,19] have been proposed to adaptively help the fusion process. The survey indicates that although pixel-level, transform-based, and hybrid methods have achieved meaningful progress, several persistent drawbacks still limit their effectiveness. Recent strategic methods, including convolutional neural networks (CNNs), attention-guided fusion strategies, and transformer architectures, represent a major step forward by automatically learning fusion rules and modelling long-range dependencies [20,21]. However, these approaches typically require large annotated datasets and are computationally expensive.

Among various strategies, medical image fusion techniques utilizing fuzzy set (FS) principles have garnered significant credit for their adeptness in mitigating uncertainty prevailing in medical images [22]. Building upon this foundation, many researchers have endeavoured to enhance outcomes by integrating intuitionistic fuzzy set (IFS) methodologies, which incorporate membership and non-membership functions, alongside a degree of hesitation [23]. Extensive literature attests to the ongoing efforts to refine fusion outcomes through the synergistic integration of FS or IFS methodologies with other fusion techniques [24,25]. In 2017, Soundrapandiyan et al. introduced a novel methodology for fusing CT and MRI images, employing DWT in conjunction with IFS techniques (DWT-IFS) [26]. The process commences with the fusion of the two images utilizing the DWT method, then converting the resultant fused image into an intuitionistic fuzzy image (IFI) by applying membership and non-membership functions inherent in IFS. Subsequently, these IFIs are fused using the intersection and union operations within the framework of IFS. Further, in 2018, Kumar et al. contributed to enhancing IFS fusion techniques for medical images (IIFS) [27]. Their approach involved the initial application of filters as a pre-processing step for contrast enhancement, followed by conversion into the HSI model to facilitate IFI transformation. Finally, the fusion process incorporated a fusion rule based on counting whiteness and blackness to fuse the images effectively. Moreover, a multi-modal medical image fusion approach that uses the strengths of the non-subsampled contourlet transform (NSCT) in combination with fuzzy entropy was developed by Li et al. in the year 2020 (NSCT-FE) [28]. Initially, the image undergoes decomposition into low and high frequency sub-bands using NSCT. The membership degree of low-frequency coefficients is computed, while fuzzy entropy is utilized to guide coefficient fusion. High-frequency components are fused by maximizing regional energy, and the inverse transformation yields the fused image. Thereafter, in 2021, Wang et al. developed a fusion approach that initially employs Gabor filter banks to filter CT and MR image sets, yielding distinct Gabor representations for each image pair [29]. These representations train corresponding CNNs, generating Gabor-CNNs (G-CNNs). Each set is inputted into the respective trained G-CNN for introductory integration. Subsequently, a fuzzy neural network is utilized to fuse numerous results from the G-CNNs, yielding the fused result (G-CNN-FNN). Similarly, in 2023, Haribabu and Guruviah proposed a fusion technique for medical images utilizing IFS and fuzzy cross-correlation (IFS-FCC) [30]. Initially, the source images underwent conversion into IFI and were subsequently decomposed into blocks. These blocks were fused using values derived from fuzzy cross-correlation.

Primarily, three common challenges arise in fusion techniques utilizing FS and IFS. The first challenge is the effective enhancement of low-illuminated regions in brain images, where subtle lesions are often difficult to detect [31]. The second challenge concerns the selection of an appropriate intuitionistic fuzzy generator (IFG). Fusion performance is highly sensitive to the choice of membership and non-membership functions, and an unsuitable IFG may lead to poor results [32]. The third challenge lies in fusing images in a way that preserves essential features while avoiding redundancy [33]. This paper collectively addresses these three critical issues, offering a comprehensive solution to these challenges. To overcome the first challenge, Nakagami imaging (NI) based on the Nakagami distribution is employed [34]. NI enhances poorly illuminated regions and highlights suspicious areas even when the contrast between tissues is minimal. The utilization of Nakagami distribution in ultrasound signal analysis was pioneered by Shankar in 2000 [35], while in recent years, works by Alphar extended its application to MRI brain images, successfully identifying lesions [36,37]. The second challenge is tackled by introducing a novel IFG specifically designed for fusion. Our earlier work proposed an IFG for clustering brain images and achieved strong segmentation accuracy [38]. In this study, a new IFG is developed to reduce uncertainty and enrich spatial information in IFIs. Finally, the third challenge is addressed by extracting contrast features using gray-level co-occurrence matrix (GLCM) analysis [39]. These features help retain crucial structural details while filtering redundant information, thereby preserving fine image details. Overall, Table 1 presents a summary of existing shortcomings, the corresponding research challenges, and the targeted solutions offered by the proposed method.

Based on the preceding analysis, this study introduces a novel fusion method termed Nakagami distribution and intuitionistic fuzzy set-based multi-modal brain image fusion (NIFMBF). Initially, brain images from diverse modalities undergo NI in both vertical and horizontal directions to detect and differentiate doubtful regions. Then, the outputs of NI are mapped into RGB channels to maximize detail preservation. The use of RGB mapping provides an intuitive and functionally meaningful separation of diagnostic information. By assigning NI outputs to three distinct colour channels, subtle lesion-enhanced regions (red), non-lesion structural areas (blue), and redundant or background regions (green) are independently processed. This channel-wise decomposition prevents dominant structures from suppressing weaker features and ensures that each category of information contributes proportionally during fusion. Subsequently, the resulting RGB colour channels of each

Table 1
Mapping of limitations, research challenges, and proposed solutions.

Limitations of existing studies	Research challenges	Proposed solutions
Spatial-domain methods suppress fine details, introduce noise, and cause spectral distortion.	Difficulty in enhancing low-illumination regions and revealing imperceptible lesions.	NI statistically models intensity distribution, enhances contrast in poorly illuminated regions, and reveals subtle abnormalities.
IFS-based approaches depend heavily on the design of membership/non-membership functions.	Uncertainty in selecting or designing an appropriate IFG.	A novel IFG is developed to construct reliable IFI, which reduces vagueness and improves detail representation.
Multi-scale and multi-resolution methods preserve edges but often suffer from gray value degradation and redundancy of features.	Preserving essential features while avoiding redundancy in the fusion process.	GLCM-based contrast feature extraction captures structural details that guides block-wise fusion, and filters out redundant information.

NI undergo IFS transformation using a newly developed IFG to generate enhanced IFIs. This step facilitates the following fusion processes by producing enriched IFIs that preserve complementary structural and contrast information across channels. Next, GLCM-based contrast features are extracted from the IFI of each channel to capture fine details. These features guide the fusion of the three IFIs. Finally, the fused images from the three channels are concatenated to produce the complete fused image. Furthermore, to validate the proposed NIFMBF method, an extensive experimental analysis is conducted using three distinct datasets, benchmarked against five other state-of-the-art fusion methods. The significant contributions of the proposed NIFMBF technique are given as follows:

(i) The proposed method utilizes NI to effectively expose imperceptible brain lesions and subtle abnormalities, even in regions where the contrast between tissues is minimal.
(ii) The NI process generates three separate channels, and each is independently enhanced; this process ensures that the contrast improvements in one channel complement and reinforce those in the others, resulting in overall image clarity.
(iii) A novel IFG is developed specifically for brain image fusion, which enhances fine details within the three NI channels and optimizes them for the subsequent fusion process.

The paper begins by establishing the foundational framework in Section 2, followed by a detailed explanation of the methodology for the proposed fusion method in Section 3. Section 4 provides an extensive experimental analysis, while Section 5 discusses the outcomes derived from the experimental analysis. The conclusion and future directions of the work are presented in Section 6.

2. Theoretical foundation

This section delves into fundamental definitions and conceptual frameworks essential to the scope of this work.

2.1. Intuitionistic fuzzy generator

In real-world scenarios, certain situations emerge where FS struggles to effectively address ambiguities, leading to uncertainty in assigning membership functions. To mitigate this uncertainty, a hesitation is introduced in defining the membership function within FS. Consequently, IFS extends beyond FS by collectively encompassing membership, non-membership, and hesitation degrees. This approach enables IFS to tackle numerous practical challenges effectively.

Definition 1. An IFS [40] with intuitionistic fuzzy index for a finite set $U = \{u_1, u_2, u_3, \dots, u_n\}$ is mathematically formulated as

$$S = \{(u, \mu_S(u), \nu_S(u), \pi_S(u)) | u \in U\}, \tag{1}$$

where the functions $\mu_S(u), \nu_S(u), \pi_S(u) : U \to [0,1]$ assigns the membership, non-membership and hesitation values to each element $u \in U$ within S, and subject to the constraint:

$$\mu_S(u) + \nu_S(u) + \pi_S(u) = 1. \tag{2}$$

Definition 2. A function $\Xi : [0,1] \to [0,1]$ is termed as IFG [41] if

$$\Xi(u) \le 1 - u \quad \forall u \in [0,1], \tag{3}$$

with $\Xi(0) \le 1$ and $\Xi(1) = 0$.

An IFG serves as a tool to derive IFS from FS. While an IFG can be constructed utilizing the fuzzy complement, it is essential to note that not all fuzzy complements constitute an IFG; specific conditions must be adhered to for its formation. This will be discussed in the following section.

2.2. Nakagami distribution

The Nakagami distribution is a fundamental probability distribution function (PDF) that characterises random signals. It provides a robust mathematical framework for analysing intensity distribution functions. NI utilizes this statistical distribution to detect suspicious regions within medical images that may not be discernible to the human eye. This imaging technique proves reliable even in scenarios with minimal contrast between regions, allowing for the identification of imperceptible lesions and the differentiation of cysts, fat, and tumor components within medical images.

NI relies on the Nakagami distribution [34], which is characterised by the PDF presented below:

$$f(u, m, \Omega) = \frac{2m^m u^{2m-1}}{\Gamma(m)\Omega^m} exp\left(-\frac{m}{\Omega}u^2\right), \tag{4}$$

where $\Gamma(m)$ denotes the Gamma function, and u represents the pixels of an image U. This distribution is dependent on two parameters: the scaling parameter Ω and the m-parameter, which are expressed in Eqs. (5) and (6), respectively, as

$$\Omega = M(U^2), \tag{5}$$

$$m = \frac{\Omega^2}{M[(U^2 - \Omega)^2]}. \tag{6}$$

Here, M represents the statistical mean, and both parameters vary according to the intensity of the image. When $m = 1$, the function corresponds to the Rayleigh distribution; for $m < 1$, it transforms into a pre-Rayleigh distribution, and for $m > 1$, it becomes a post-Rayleigh distribution. Consequently, by adjusting the value of m, various characteristics of the image become apparent.

2.3. Texture feature extraction

Typically, texture features encapsulate the structure and characteristics of surfaces. Integral to image fusion techniques, texture feature extraction serves to identify crucial image attributes and facilitate appropriate image integration. This extraction process is categorised into spatial and spectral texture feature extraction methods. Among these, the GLCM is a prominent spatial texture feature extraction technique that uses pixel statistics. GLCM computation defines its dimensions based on the number of gray levels in the image [39]. For an image U with g distinct intensity levels, the co-occurrence matrix B of size $g \times g$ is formed over the $a \times b$ image region, as shown in Eq. (7).

$$B_{\Delta p, \Delta q}(r,t) = \sum_{p=1}^{a}\sum_{q=1}^{b} \begin{cases} 1, & U(p,q) = r \text{ and } U(p+\Delta p, q+\Delta q) = t, \\ 0, & otherwise. \end{cases} \tag{7}$$

where r and t represent the corresponding pixels, p and q denote the positions of the pixels and $(\Delta p, \Delta q)$ is the interspace between the pixel p and corresponding neighbouring pixel q. Several texture features such as energy, contrast, correlation, and homogeneity can be computed from the GLCM. In this study, the contrast feature is utilized to capture the variation in intensity levels across the image, which helps in highlighting structural differences and edge information. This feature is extracted from the image and is mathematically expressed as follows:

$$C = \frac{1}{g \times g} \sum_r \sum_t (r-t)^2 B_{\Delta p, \Delta q}(r,t). \tag{8}$$

Thus, the contrast feature utilizing GLCM is extracted from the image and elaborated upon in detail in the subsequent section.

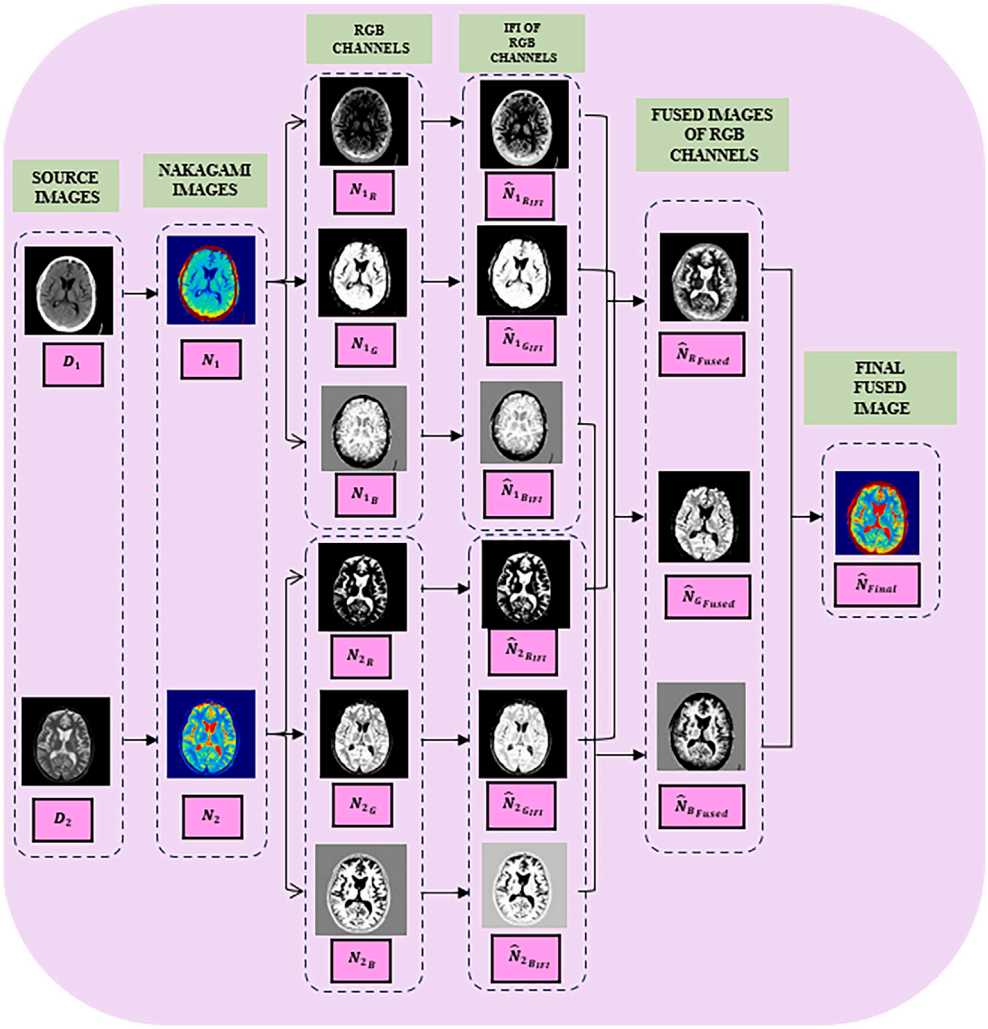


Fig. 1. The flow chart of the proposed NIFMBF fusion algorithm.

3. Proposed methodology

The proposed methodology unfolds in two distinct sections. Initially, it examines the evolutionary process of the innovative IFG for brain image fusion. Subsequently, it elucidates the development of the proposed NIFMBF method, designed to integrate multi-modal medical images seamlessly.

3.1. Evolution of IFG

This section focuses on the development of the IFG. Utilizing the IFS proves to be a more practical approach in handling uncertainty within brain images, and the IFG plays a pivotal role in constructing IFS from the FS. Therefore, careful selection of the IFG is paramount as it directly impacts the problem at hand. Here, the IFG is framed explicitly for brain image fusion. To commence, a fuzzy complement function is devised, utilizing an increasing function that ensures a monotonic and consistent representation of non-membership, which stabilizes the construction of IFIs. It is outlined as follows.

$$f(u) = \frac{1}{\beta} log\left[1 + u\left(1 + e^{\beta}\right)\right], \tag{9}$$

such that $f(0) = \frac{1}{\beta} log[1] = 0$ and β is a constant. Then, an involutive fuzzy complement ξ can be considered if there exist a function $\xi : [0,1] \to [0,1]$ such that $\xi(u) = f^{-1}[f(1) - f(u)]$. Accordingly, the inverse of this function is given as below

$$g^{-1}(w) = \frac{e^{\beta w} - 1}{1 + e^{\beta}}, \tag{10}$$

On substituting the values, the equation becomes

$$\begin{aligned}\xi(u) &= f^{-1}\left[\frac{1}{\beta} log\left(2 + e^{\beta}\right) - \frac{1}{\beta} log\left(1 + u\left(1 + e^{\beta}\right)\right)\right] \\ &= f^{-1}\left[\frac{1}{\beta} log\left(\frac{2 + e^{\beta}}{1 + u\left(1 + e^{\beta}\right)}\right)\right].\end{aligned} \tag{11}$$

On solving Eq. (11), it is obtained as

$$\xi(u) = \frac{1 - u}{1 + u\left(1 + e^{\beta}\right)}. \tag{12}$$

If the membership function $\mu(u)$ is considered, then the non-membership function is obtained as in Eq. (13).

$$\xi(\mu(u)) = \frac{1 - \mu(u)}{1 + \mu(u)\left(1 + e^{\beta}\right)}. \tag{13}$$

Eq. (13) constitutes a fuzzy complement. However, it is essential to recognize that not all fuzzy complements qualify as IFG; they must adhere to the conditions outlined in Definition 2. It is worth highlighting

that Eq. (13) satisfies the conditions specified in Definition 2 for any value of β. Consequently, the development of IFG is established and it is denoted as:

$$\Xi(\mu(u)) = \frac{1-\mu(u)}{1+\mu(u)\left(1+e^{\beta}\right)}. \tag{14}$$

3.2. The proposed NIFMBF methodology

The NIFMBF method aims to perfectly merge multi-modal medical images, including those with low illumination and uncertainty, into a single image containing extensive complementary information with high contrast. The schematic representation of the proposed NIFMBF fusion method is displayed in Fig. 1. The proposed NIFMBF methodology is presented in well-structured subsections, where each step of the process is explained in detail for clarity.

3.2.1. Nakagami imaging

The NIFMBF method ensures that all required features are extracted and fused effectively into a unified image, even when encountering regions with minimal contrast differentiation. To achieve this, the source images first undergo the NI process. NI enhances the visibility of subtle lesions while suppressing less relevant regions. This helps to differentiate the various brain structures. Below, the outlined procedure details the transformation of source images into NI.

Consider two source images, denoted as $\mathcal{D}_i$ where $i = 1, 2$, each with dimensions $K \times J$ and pre-registered with L levels of grayness. These images undergo conversion into NI by applying the Nakagami distribution function, as elaborated in Section 2.2. This distribution function is applied to both the rows and columns of the source images. It is delineated for a single source image to elucidate the conversion process.

The matrix form of the considered source image is as follows

$$\mathcal{D} = \begin{bmatrix} d(1,1) & d(1,2) & \dots & d(1,j) \\ d(2,1) & d(2,2) & \dots & d(2,j) \\ \vdots & \vdots & \ddots & \vdots \\ d(k,1) & d(k,2) & \dots & d(k,j) \end{bmatrix}, \tag{15}$$

where $d(x, y)$ notates the intensity of the pixel at the position (x, y). The pixel values are vectorized into an array by using Eq. (16).

$$\begin{aligned} \mathcal{V} &= [d(1,1) \quad d(1,2) \quad \dots \quad d(k,j)] \\ &= [v_1 \quad v_2 \quad \dots \quad v_Q], \end{aligned} \tag{16}$$

such that $Q = kj$. Now, to access the column pixel values of the matrix $\mathcal{D}$, it is transposed. Thus, the vector notion is obtained as in Eq. (18).

$$\mathcal{D}^T = \begin{bmatrix} d(1,1) & d(2,1) & \dots & d(k,1) \\ d(1,2) & d(2,2) & \dots & d(k,2) \\ \vdots & \vdots & \ddots & \vdots \\ d(1,j) & d(2,j) & \dots & d(k,j) \end{bmatrix}, \tag{17}$$

$$\begin{aligned} \bar{\mathcal{V}} &= [d(1,1) \quad d(2,1) \quad \dots \quad d(k,j)] \\ &= [\bar{v}_1 \quad \bar{v}_2 \quad \dots \quad \bar{v}_Q]. \end{aligned} \tag{18}$$

Subsequently, in line with the prior discussion in Eq. (5), the scaling parameter Ω is determined by computing the statistical mean. Given that the statistical mean for $\mathcal{V}$ and $\bar{\mathcal{V}}$ is identical, the value of Ω can be computed using either vector.

$$\Omega = \frac{1}{Q}\sum_{a=1}^{Q} v_a. \tag{19}$$

Similarly, the Nakagami m-parameter can be computed using $\mathcal{V}$ or $\bar{\mathcal{V}}$ and it is represented as

$$m = \frac{\Omega^2}{\sum_{a=1}^{Q}\left(v_a^2 - \Omega\right)^2}. \tag{20}$$

Following this, the updated intensity of the image is derived by implementing the Nakagami distribution in both row and column directions, as expressed in Eqs. (21) and (22).

$$\tilde{v}_a \in \tilde{\mathcal{V}}_Q = \frac{2m^m v_a^{2m-1}}{\Gamma(m)\Omega^m} exp\left(-\frac{m}{\Omega}v_a^2\right), \quad a = 1, 2, \dots, Q, \tag{21}$$

$$\check{v}_a \in \check{\mathcal{V}}_Q = \frac{2m^m \bar{v}_a^{2m-1}}{\Gamma(m)\Omega^m} exp\left(-\frac{m}{\Omega}\bar{v}_a^2\right), \quad a = 1, 2, \dots, Q. \tag{22}$$

The obtained updated pixel values are then reconstructed to their original positions according to Eqs. (23) and (25) and reshaped into matrices as described in Eqs. (24) and (26).

$$\begin{aligned} \tilde{\mathcal{V}}_Q &= [\tilde{v}_1 \quad \tilde{v}_2 \quad \dots \quad \tilde{v}_Q] \\ &= [\tilde{d}(1,1) \quad \tilde{d}(1,2) \quad \dots \quad \tilde{d}(k,j)], \end{aligned} \tag{23}$$

$$\tilde{\mathcal{D}} = \begin{bmatrix} \tilde{d}(1,1) & \tilde{d}(1,2) & \dots & \tilde{d}(1,j) \\ \tilde{d}(2,1) & \tilde{d}(2,2) & \dots & \tilde{d}(2,j) \\ \vdots & \vdots & \ddots & \vdots \\ \tilde{d}(k,1) & \tilde{d}(k,2) & \dots & \tilde{d}(k,j) \end{bmatrix}, \tag{24}$$

$$\begin{aligned} \check{\mathcal{V}}_Q &= [\check{v}_1 \quad \check{v}_2 \quad \dots \quad \check{v}_Q] \\ &= [\check{d}(1,1) \quad \check{d}(2,1) \quad \dots \quad \check{d}(k,j)], \end{aligned} \tag{25}$$

$$\check{\mathcal{D}} = \begin{bmatrix} \check{d}(1,1) & \check{d}(2,1) & \dots & \check{d}(k,1) \\ \check{d}(1,2) & \check{d}(2,2) & \dots & \check{d}(k,2) \\ \vdots & \vdots & \ddots & \vdots \\ \check{d}(1,j) & \check{d}(2,j) & \dots & \check{d}(k,j) \end{bmatrix}, \tag{26}$$

Finally, the NI is obtained for the considered source image by the dot product of the reconstructed matrices and it is represented as follows

$$\mathcal{N} = \tilde{\mathcal{D}}.\check{\mathcal{D}}. \tag{27}$$

Accordingly, applying the procedure in Eqs. (15)–(27) yields the Nakagami images $\mathcal{N}_i$, where $i = 1, 2$ denotes the two source images. This transformation models the statistical behaviour of ultrasound signals, which allows subtle variations in tissue texture to be amplified. Consequently, each NI is colorized to highlight the variation between different brain regions and denoted as $\mathcal{N}_z$ such that $z = r, g, b$ represents the RGB channels. In this representation, the red channel of the NI signifies lesions or suspicious regions, while the blue channel represents areas excluding the lesions. Meanwhile, the green channel indicates redundant areas within the brain. To visually illustrate these distinctions, all three channels for a sample image are depicted in Fig. 1.

3.2.2. Obtaining IFI through the developed IFG

As a subsequent process, the individual channels of the NI undergo enhancement through the entropy-based enhancement method of IFS. Entropy-based enhancement is utilized because the parameter β strongly influences the resulting IFI. Different values of β produce varying degrees of structural and contrast detail. Intuitively, β acts as a control factor that adjusts the balance between membership and non-membership information, thereby influencing the sensitivity of the IFI to image details. The entropy quantifies the amount of uncertainty or information present in the fuzzy representation. By adjusting β to maximize entropy, the system emphasizes features that carry the highest information content, which enhances the visibility of fine details in the IFIs. This process ensures that the generated IFIs retain adequate structural and contrast information for fusion. In accordance with this, the developed IFG is employed to derive the intuitionistic fuzzy image IFI. Initially, the membership function of fuzzy image (FI) for each channel is obtained by normalizing each channel of NI, as illustrated below:

$$\mu_{FI}(\mathcal{N}_z(k,j)) = \frac{\mathcal{N}_z(k,j) - \mathcal{N}_{z_{min}}}{\mathcal{N}_{z_{max}} - \mathcal{N}_{z_{min}}}, \tag{28}$$

where $\mathcal{N}_z(k, j)$ denotes the intensity of the respective channel z of NI at $(k, j)^{th}$ location. Also, the $\mathcal{N}_{z_{min}}$ and $\mathcal{N}_{z_{max}}$ denote the minimum and

maximum pixel values of the corresponding $\mathcal{N}_z$. Further, the FI is converted into IFI by using the IFG obtained in Eq. (14). The proposed IFG reduces the dependency and provides a more adaptive mapping suited to brain image fusion thereby improving robustness and reliability. The membership function of IFI is expressed as in Eq. (29).

$$\mu_{IFI}(\mathcal{N}_z(k,j);\beta) = 1 - \frac{1-\mu_{FI}(\mathcal{N}_z(k,j))}{1+\mu_{FI}(\mathcal{N}_z(k,j))\left(1+e^{\beta}\right)} = \frac{\left(e^{\beta}+2\right)\mu_{FI}(\mathcal{N}_z(k,j))}{1+\mu_{FI}(\mathcal{N}_z(k,j))\left(1+e^{\beta}\right)}. \tag{29}$$

To obtain the non-membership function, the negation function given in Eq. (12) is employed and β is changed into $\beta+1$. Thus, the non-membership function of IFI is derived as follows

$$\nu_{IFI}(\mathcal{N}_z(k,j);\beta) = \frac{1-\mu_{FI}(\mathcal{N}_z(k,j))}{1+(2e^{\beta}+e^{2\beta+1}+2e^{\beta+1}+3)\mu_{FI}(\mathcal{N}_z(k,j))}. \tag{30}$$

Finally, the hesitation degree of IFI is incurred as in Eq.. (31).

$$\pi_{IFI}(\mathcal{N}_z(k,j);\beta) = 1-\mu_{IFI}(\mathcal{N}_z(k,j);\beta)-\nu_{IFI}(\mathcal{N}_z(k,j);\beta). \tag{31}$$

All these degrees consist of a parameter β and images change according to each value of β. Thus, the optimum value of β is obtained using the intuitionistic fuzzy entropy [42].

$$ENT_{IFI} = \frac{1}{K\times J}\sum_{i=0}^{K-1}\sum_{j=0}^{J-1} \times \frac{2\mu_{IFI}(\mathcal{N}_z(k,j);\beta)\nu_{IFI}(\mathcal{N}_z(k,j);\beta)+\pi^2_{IFI}(\mathcal{N}_z(k,j);\beta)}{\mu^2_{IFI}(\mathcal{N}_z(k,j);\beta)+\nu^2_{IFI}(\mathcal{N}_z(k,j);\beta)+\pi^2_{IFI}(\mathcal{N}_z(k,j);\beta)}. \tag{32}$$

Therefore, the optimum value of β corresponds to the maximum value of the entropy. Therefore, all the degrees are acquired using this optimum value. This entropy formula adjusts the contrast based on the amount of information present in the local neighbourhood. It ensures that areas rich in structural details are enhanced, while flat regions remain smooth. On the whole, the IFI for each channel is represented as given below

$$\hat{\mathcal{N}}_{z_{IFI}} = \Big\{\mathcal{N}_z(k,j),\ \mu_{IFI}(\mathcal{N}_z(k,j);\beta),\ \nu_{IFI}(\mathcal{N}_z(k,j);\beta), \pi_{IFI}(\mathcal{N}_z(k,j);\beta)\ \Big|\ \mathcal{N}_z(k,j)\in\{0,1,\ldots,L-1\}\Big\}. \tag{33}$$

The IFI quantifies the dependence with which a pixel is classified as belonging to an informative region rather than a redundant one. A higher IFI value suggests that stronger structural details should be preserved during the fusion process.

3.2.3. GLCM-based contrast feature extraction

Now, the IFI for each channel of the NI is forwarded to the subsequent NIFMBF fusion process. Textural characteristics are derived from each IFI to amalgamate all necessary information, prioritizing essential data. To achieve this, each IFI is divided into blocks of size $f\times h$ for localized processing to extract features such that $\hat{\mathcal{N}}_{z_o}$ represents the oth block of $\hat{\mathcal{N}}_z$. This block size was chosen to balance detail preservation with computational efficiency: smaller blocks capture local variations more accurately but increase processing time, while larger blocks reduce complexity at the cost of fine detail. Contrast features, utilizing GLCM, are extracted from these IFI blocks through the equations delineated in Eqs. (7) and (8). This helps to capture second-order pixel dependencies, which are critical for highlighting structural and contrast details in medical images. Furthermore, it provides a more comprehensive representation of local texture patterns relevant for accurate brain image fusion. The procedures mentioned above have been executed for the second source image. As a result, contrast features have been extracted from blocks within each IFI across the three channels of the NI for the two source images.

Algorithm 1 Flow of proposed NIFMBF algorithm.

Require: Two source images D_i, i=1,2 with $K\times J$ dimensions, size of the blocks $f\times h$.

Step 1: Utilize the Nakagami distribution to convert the source images, $\mathcal{D}_1$ and $\mathcal{D}_2$, into NI by applying the corresponding equations from Eq. (15) to Eq. (27) in both vertical and horizontal directions.

Step 2: Employ the developed IFG to transform the three channels of NI into IFI, referencing the equations from Eq. (28) to Eq. (33).

Step 3: Divide each IFI into blocks of dimensions $f\times h$.

Step 4: Compute the contrast feature for each block of IFIs using the GLCM as specified in Eqs. (7) and (8).

Step 5: Merge the blocks of three channels for the two IFIs into single blocks using the fusion rule delineated in Eq. (34).

Step 6: Reconstruct the fused blocks of each channel into matrices of dimensions $K\times J$.

Step 7: Defuzzify each of the three channels into a crisp image according to the representation in Eq. (35).

Step 8: Concatenate all three channels into a single fused output image.

3.2.4. Channel-wise fusion and reconstruction

The fusion process is initiated by blending blocks from each channel of NI for the two source images, utilizing the extracted contrast features. The fusion rule is outlined as follows.

$$\hat{\mathcal{N}}_{z_{fo}} = \begin{cases} \hat{\mathcal{N}}_{1_{z_o}}(k,j), & if \quad Contrast_{1_{z_o}} > Contrast_{2_{z_o}} \\ \hat{\mathcal{N}}_{2_{z_o}}(k,j), & if \quad Contrast_{1_{z_o}} < Contrast_{2_{z_o}} \\ \frac{\hat{\mathcal{N}}_{1_{z_o}}(k,j)+\hat{\mathcal{N}}_{2_{z_o}}(k,j)}{2}, & Otherwise \end{cases} \tag{34}$$

After that, the blocks from each channel are reconstructed into an image denoted as $\hat{\mathcal{N}}_{z_{Fused}}$. Following this, a defuzzification process is applied to obtain crisp images from IFIs for each fused channel. This ensures that the uncertainty modelled in the intuitionistic fuzzy domain is translated into practical quantitative measures that can be directly used in subsequent computations. It is outlined as in Eq. (35).

$$\hat{\mathcal{N}}_{z_f} = (\mathcal{N}_{z_{max}} - \mathcal{N}_{z_{min}}) * \hat{\mathcal{N}}_{z_{Fused}} + \mathcal{N}_{z_{min}} \tag{35}$$

Further, each defuzzified image of three channels is concatenated together as $\hat{\mathcal{N}}_{Final}$ to form a final fused brain image. This fusion rule combines the complementary information from all processed channels. It ensures that the most informative pixel values from each source are integrated into the final fused output. The whole process of NIFMBF method is scrutinized in the Algorithm 1.

4. Experimental results and analysis

4.1. Experimental configuration

To ensure comprehensive experimental analysis, four distinct types of databases were assembled from publicly available datasets, each showcasing variations in brain diseases, imaging modalities, and dimensions. The selection aimed to capture a diverse range of scenarios encountered in clinical practice. The first two databases were sourced from the whole brain atlas provided by Harvard University [43]. The first database comprises CT and MRI modalities of 256×256 dimensions. Meanwhile, the second database includes GAD-MRI and T2-weighted MRI images of 256×256 dimensions. Moving on, the third database contains T1-weighted MRI and T2-weighted MRI images, each with dimensions of 256 × 256, sourced from a multiple sclerosis (MS) dataset [44].

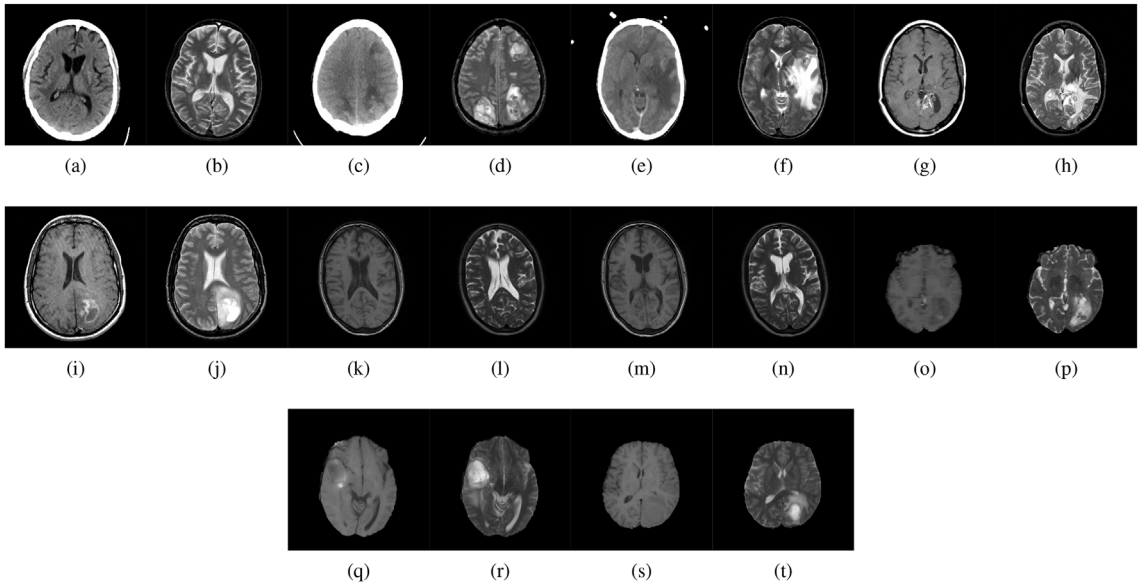


Fig. 2. Source images:(a-f)-Database-I:(a,b)-Pair-1, (c,d)-Pair-2, (e,f)-Pair-3; (g-j)-Database-II:(g,h)-Pair-4, (i,j)-Pair-5; (k-n)-Database-III:(k,l)-Pair-6, (m,n)-Pair-7; (o-t)-Database-IV:(o,p)-Pair-8, (q,r)-Pair-9, (s,t)-Pair-10.

Lastly, the fourth database comprises T1-weighted and T2-weighted MRI images with dimensions of 240×240, obtained from the BraTS dataset, focusing on glioma brain tumors [45]. The utilization of these diverse datasets facilitates a thorough investigation and a comprehensive evaluation of the proposed fusion technique across different pathological conditions.

In order to rigorously evaluate the performance of the proposed fusion technique, it is imperative to compare its results with those obtained from other benchmarking state-of-the-art fusion methods. Given that the proposed fusion method primarily focuses on mitigating uncertainties in brain images using IFS, comparative methods from the same domain in recent years are selected. At the same time, to ensure broader relevance, the comparison also includes representative methods from other categories, such as multi-resolution transforms, multi-scale transform methods, and learning-driven frameworks. This allows the advantages of the proposed method to be demonstrated not only within the IFS field but also across a wider range of image fusion paradigms. Consequently, five other fusion methods have been chosen for result comparison: DWT-IFS [26], IIFS [27], NSCT-FE [28], G-CNN-FNN [29], and IFS-FCC [30]. Detailed information regarding these methods has been provided in the introduction 1. Throughout the comparison process, the parameters utilized in all these methods have been maintained at the values specified in their respective papers to ensure consistency and fairness in the evaluation. Furthermore, a 2×2 block size was selected as it effectively preserves local details while maintaining computational efficiency. Accordingly, the experiments were conducted for the proposed fusion method and the five fusion methods mentioned above across four types of databases, employing MATLAB 2019a software for implementation. A larger number of input images from these databases were subjected to testing, and a subset comprising ten pairs of images was selected for presentation in the paper to maintain clarity and conciseness. The ten pairs of source images were chosen to be representative of the larger dataset, providing a balanced and fair evaluation, and they are depicted in Fig. 2.

The statistical metrics are used to evaluate the results generated by the proposed NIFMBF method and other comparative fusion methods. Five fusion metrics have been employed within this framework to assess various aspects of the fused images. These metrics are spatial frequency (SF), entropy (EN), universal image quality index (UIQI), average gradient (AG), and mutual information (MI). Each of these metrics quantifies specific characteristics of an image, providing valuable insights into its quality and fidelity. The formula for each metric is detailed as follows:

1. The SF of an image indicates the scale of information and intensity variations within the image [23]. A higher SF value typically corresponds to a better fusion outcome, as it signifies a greater abundance of information and nuanced intensity changes. The SF is mathematically defined as

$$\begin{aligned} SF &= \sqrt{RF^2 + CF^2}, \\ RF &= \sqrt{\frac{1}{K \times J} \sum_{k=1}^{K} \sum_{j=2}^{J} \left(\hat{\mathcal{N}}_{Final}(k,j) - \hat{\mathcal{N}}_{Final}(k,j-1)\right)^2}, \\ CF &= \sqrt{\frac{1}{K \times J} \sum_{k=2}^{K} \sum_{j=1}^{J} \left(\hat{\mathcal{N}}_{Final}(k,j) - \hat{\mathcal{N}}_{Final}(k-1,j)\right)^2}. \end{aligned} \tag{36}$$

   where CF and RF represent column and row frequencies of an image.
2. The EN of an image quantifies the level of knowledge present in the fused image. Higher EN values indicate a more significant amount of information and complexity, which typically correspond to improved fusion results. The EN is formulated as

$$EN = -\sum_{i=0}^{L-1} (prob_i \ \log prob_i) \tag{37}$$

   where $prob_i$ denotes the probability of the occurrences of gray level i in the fused image $\hat{\mathcal{N}}_{Final}$.
3. The UIQI metric combines three key aspects: correlation, luminance, and contrast, to assess the similarity between the fused and source images [46]. A higher UIQI value signifies a closer resemblance between the fused output and the original images, indicating minimal distortion. The UIQI is defined as follows

$$UIQI(\mathcal{D}_1, \mathcal{D}_2, \hat{\mathcal{N}}_{Final}) = \frac{UIQI(\mathcal{D}_1, \hat{\mathcal{N}}_{Final}) + UIQI(\mathcal{D}_2, \hat{\mathcal{N}}_{Final})}{2} \tag{38}$$

where,

$$UIQI(D_1,\hat{\mathcal{N}}_{Final}) = \frac{\sigma_{D_1\hat{\mathcal{N}}_{Final}}}{\sigma_{D_1}\sigma_{\hat{\mathcal{N}}_{Final}}}\cdot\frac{2\overline{D_1\hat{\mathcal{N}}_{Final}}}{\bar{D_1}^2+\overline{\hat{\mathcal{N}}}^2_{Final}}\cdot\frac{\sigma_{D_1}\sigma_{\hat{\mathcal{N}}_{Final}}}{\sigma^2_{D_1}+\sigma^2_{\hat{\mathcal{N}}_{Final}}}$$

$$UIQI(D_2,\hat{\mathcal{N}}_{Final}) = \frac{\sigma_{D_2\hat{\mathcal{N}}_{Final}}}{\sigma_{D_2}\sigma_{\hat{\mathcal{N}}_{Final}}}\cdot\frac{2\overline{D_2\hat{\mathcal{N}}_{Final}}}{\bar{D_2}^2+\overline{\hat{\mathcal{N}}}^2_{Final}}\cdot\frac{\sigma_{D_2}\sigma_{\hat{\mathcal{N}}_{Final}}}{\sigma^2_{D_2}+\sigma^2_{\hat{\mathcal{N}}_{Final}}}$$

where $\bar{D_1},\bar{D_2},\overline{\hat{\mathcal{N}}}_{Final}$ denote mean of $D_1, D_2, \hat{\mathcal{N}}_{Final}$ respectively. $\sigma^2_{D_1},\sigma^2_{D_2},\sigma^2_{\hat{\mathcal{N}}_{Final}}$ indicate the variance of $D_1, D_2, \hat{\mathcal{N}}_{Final}$ respectively. $\sigma_{D_1\hat{\mathcal{N}}_{Final}}$ is the covariance between D_1 and $\hat{\mathcal{N}}_{Final}$.

4. The AG measures the sharpness of an image while also assessing edge preservation [47]. A higher AG value indicates sharper transitions between different regions in the fused image, contributing to better overall quality. The AG is outlined as

$$AG = \frac{1}{K\times J}\sum_{k=1}^{K}\sum_{j=1}^{J}\Big[\Big(\hat{\mathcal{N}}_{Final}(k,j)-\hat{\mathcal{N}}_{Final}(k+1,j)\Big)^2 + \Big(\hat{\mathcal{N}}_{Final}(k,j)-\hat{\mathcal{N}}_{Final}(k,j+1)\Big)^2\Big]^{1/2}. \tag{39}$$

5. The MI quantifies the level of dependency between source and fused images, reflecting the amount of information transferred from the source images to the fused image. A higher MI value indicates a stronger relationship between the original and fused images, suggesting a better fusion outcome. The MI is expressed mathematically as

$$\begin{aligned} MI(D_1,D_2,\hat{\mathcal{N}}_{Final}) &= \frac{MI(D_1,\hat{\mathcal{N}}_{Final})+MI(D_2,\hat{\mathcal{N}}_{Final})}{2},\\ MI(D_1,\hat{\mathcal{N}}_{Final}) &= \sum_{k=1}^{K}\sum_{j=1}^{J} p_{D_1\hat{\mathcal{N}}_{Final}}(k,j)\log\frac{p_{D_1\hat{\mathcal{N}}_{Final}}(k,j)}{p_{D_1}(k)p_{\hat{\mathcal{N}}_{Final}}(j)},\\ MI(D_2,\hat{\mathcal{N}}_{Final}) &= \sum_{k=1}^{K}\sum_{j=1}^{J} p_{D_2\hat{\mathcal{N}}_{Final}}(k,j)\log\frac{p_{D_2\hat{\mathcal{N}}_{Final}}(k,j)}{p_{D_2}(k)p_{\hat{\mathcal{N}}_{Final}}(j)} \end{aligned} \tag{40}$$

where p_{D_1} and p_{D_2} indicate the marginal probability distribution of the source images. $p_{D_1\hat{\mathcal{N}}_{Final}}$ and $p_{D_2\hat{\mathcal{N}}_{Final}}$ represent normalized joint distribution.

4.2. Examination of the experiment

The research paper delves into an investigation of the experimental results obtained through the proposed method as well as other fusion methods, encompassing both qualitative and quantitative analyses. The qualitative analysis offers insights into the visual perception of the results, drawing upon human observation and expertise in interpreting brain images. This aspect provides a nuanced understanding that may not be quantifiable but is nonetheless crucial for comprehending the impact and effectiveness of the methods employed. On the other hand, the quantitative analysis relies on statistical fusion metrics results and evaluates the computational complexity associated with the different techniques. This aspect provides a more objective assessment, allowing for comparisons between the proposed method and alternative fusion techniques based on numerical data and computational efficiency. Both qualitative and quantitative aspects of the experimental results are thoroughly examined and presented in the subsequent subsections of the research paper.

4.2.1. Qualitative examination

The visual outcomes achieved through the proposed NIFMBF method and five other cutting-edge fusion methods are showcased in Fig. 3. This comparative display allows for the direct comparison of the results obtained from different fusion techniques. The work incorporates four databases, as discussed in the preceding section. A total of ten pairs from these databases are analysed in detail and categorised as follows. For instance, Pair-1, Pair-2, and Pair-3 are sourced from Database-I, while Pair-4 and Pair-5 originate from Database-II. Moving forward, Pair-6 and Pair-7 are drawn from Database-III, and Pair-8, Pair-9, and Pair-10 are extracted from Database-IV. This meticulous categorisation enables the systematic evaluation of the proposed NIFMBF method and other fusion approaches across various datasets. The qualitative assessment of the fusion results is conducted based on the organisation of these pairs according to their respective databases. By structuring the analysis in this manner, the clinicians can discern trends, strengths, and weaknesses specific to each fusion method within the context of different datasets.

The Pair-1, Pair-2, and Pair-3 are part of the Database-I, comprising images obtained from both CT and MRI modalities. Combining brain images from these two modalities offers notable benefits. CT scans highlight the dense structures of the brain, while MRI images provide detailed views of its soft tissues. Integrating both modalities produces a single image showcasing the anatomical and pathological features of the brain comprehensively. Additionally, including MRI alongside CT in diagnostics enhances spatial resolution, mitigating potential long-term patient exposure risks associated with CT scans. Upon analysing the visual results depicted in the first three rows of Fig. 3, several observations emerge regarding the performance of different fusion methods. First, the DWT-IFS method demonstrates a degradation of the gray and white matter structures within the brain. The fusion process appears to compromise the clarity and definition of these regions, potentially diminishing the diagnostic value of the resulting image. Next, the IIFS method intensifies the images significantly, resulting in the smudging of details. This excessive enhancement obscures subtle features and nuances in the original images, impairing the interpretability of the fused image. Subsequently, in the fusion result of NSCT-FE, a substantial amount of complementary information seems to be lost. This loss of information suggests that the fusion process fails to effectively integrate the distinct features captured by each modality, reducing overall image quality. Further, the fusion outcome of the G-CNN-FNN technique struggles to portray the hard structures produced by CT images accurately. This inadequacy results in losing critical anatomical details, particularly in regions where CT imaging provides crucial diagnostic information. Moreover, the IFS-FCC fusion method yields an output with some loss of anatomical structure. While the fusion process may achieve a degree of integration between CT and MRI modalities, the resulting image lacks the fidelity necessary for precise anatomical localization and characterisation. In contrast to the shortcomings observed in other fusion techniques, the proposed NIFMBF fusion method stands out by providing accurate information from both modalities. The regions of interest are highly distinguishable, indicating the successful integration of CT and MRI data without compromising anatomical clarity or pathological detail.

The Pair-4 and Pair-5 are the Database-II components, consisting of images acquired from both GAD-MRI and T2-weighted MRI modalities. GAD-MRI visualizes vascular abnormalities within the brain, highlighting altered blood–brain barrier permeability areas. On the other hand, T2-weighted MRI provides detailed structural information about brain tissues, enabling the visualization of anatomical features with high contrast. The fusion of these images enhances the ability to identify lesions within the brain. This integration minimises the likelihood of misinterpretation or confusion regarding the nature and type of detected lesions. Moreover, the fusion of GAD-MRI and T2-weighted MRI images facilitates a deeper understanding of the spatial relationship between lesions and surrounding tissues. This aspect is crucial for accurately assessing disease progression, as it enables pathologists to discern how lesions interact with and impact adjacent brain structures. Several significant observations emerge after analysing the results obtained from

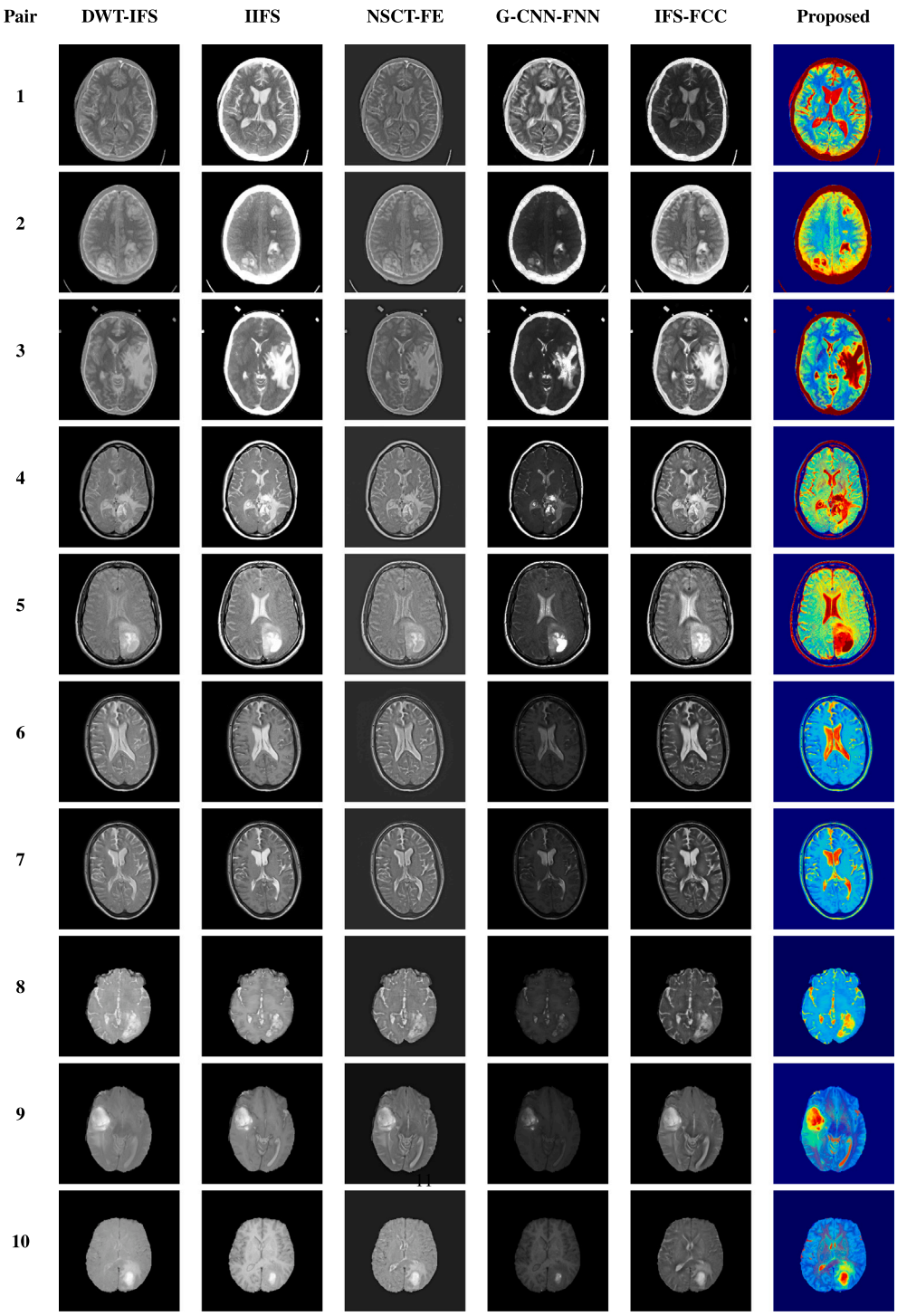


Fig. 3. Qualitative results of proposed and other benchmarking fusion methods.

Pair-4 and Pair-5 in the research study. Firstly, the DWT-IFS fusion method exhibits deteriorated fusion performance, suggesting limitations in its ability to integrate information from the modalities effectively. Subsequently, the IIFS technique demonstrates improved results, albeit with noticeable alterations in lesion size and shape attributed to over-contrast. Furthermore, the fusion outcome of NSCT-FE indicates a loss of spatial information of lesions, indicating a potential drawback of this fusion approach. Similarly, the G-CNN-FNN fused result lacks significant details from both modalities, indicating shortcomings in preserving crucial information during fusion. Moreover, the edge details are not adequately preserved in the IFS-FCC fusion method, suggesting a limitation in accurately representing fine structural features. Finally, the proposed NIFMBF fusion method successfully depicts structural changes in lesions, surrounding tissues, and anatomical structures, highlighting its effectiveness in achieving comprehensive fusion outcomes.

T1-weighted and T2-weighted MRI are utilized as two modalities for examining MS brain disease. This is considered as Database-III and is given as in Pair-6 and Pair-7. T1-weighted MRI indicates active inflammation seen in MS, whereas T2-weighted MRI is sensitive to chronic lesions, such as areas of demyelination and gliosis. Integrating images

Table 2
The values of fusion metrics for Database-I and Database-II.

Fusion results for Pair-1						
Metrics	Methods					
	DWT-IFS	IIFS	NSCT-FE	G-CNN-FNN	IFS-FCC	Proposed
SF	33.3734	83.2964	49.3815	64.7152	70.0165	**85.0719**
EN	0.7413	2.8758	1.4582	3.1341	4.1031	**4.9841**
UIQI	0.2478	0.5282	0.4681	0.6954	0.8917	**0.9532**
AG	8.2217	11.8175	9.4212	14.9743	16.2346	**18.3342**
MI	0.7629	2.1828	1.4817	3.9482	5.1123	**6.5873**
TIME	11.2187	6.8280	15.3163	35.8819	10.4173	**5.2348**
Fusion results for Pair-2						
Metrics	Methods					
	DWT-IFS	IIFS	NSCT-FE	G-CNN-FNN	IFS-FCC	Proposed
SF	33.4207	68.8242	48.3128	49.7432	53.7334	**72.9042**
EN	0.6438	2.3463	1.7765	2.9723	3.6490	**4.5298**
UIQI	0.1763	0.5789	0.3814	0.6313	0.7526	**0.8009**
AG	7.4639	9.9825	8.9347	11.0873	12.0478	**14.3873**
MI	0.8850	3.7832	1.8913	3.6382	4.6047	**5.8839**
TIME	12.3137	7.2781	16.6321	37.3342	11.4834	**6.4718**
Fusion results for Pair-3						
Metrics	Methods					
	DWT-IFS	IIFS	NSCT-FE	G-CNN-FNN	IFS-FCC	Proposed
SF	34.5418	35.7271	40.7388	53.9214	58.6821	**70.6216**
EN	0.6843	2.0043	1.7821	2.8742	3.7056	**4.8216**
UIQI	0.2327	0.5784	0.4730	0.6873	0.8113	**0.9237**
AG	8.2973	9.8871	9.2117	10.3317	11.0388	**13.4821**
MI	0.8176	2.2872	1.3218	3.7334	4.7582	**5.9659**
TIME	11.4324	6.9764	15.7482	36.3482	10.8216	**5.7032**
Fusion results for Pair-4						
Metrics	Methods					
	DWT-IFS	IIFS	NSCT-FE	G-CNN-FNN	IFS-FCC	Proposed
SF	31.7597	70.6875	40.1132	53.7877	68.8943	**72.3663**
EN	0.5835	2.1978	1.3241	2.7843	3.2494	**4.9322**
UIQI	0.2733	0.5826	0.4382	0.6105	0.8243	**0.9382**
AG	6.5140	9.4683	8.7813	12.4474	16.0349	**18.7782**
MI	0.6355	2.6828	1.3876	3.4488	4.0754	**5.8732**
TIME	9.8134	5.0078	13.4228	30.2143	9.4689	**4.8783**
Fusion results for Pair-5						
Metrics	Methods					
	DWT-IFS	IIFS	NSCT-FE	G-CNN-FNN	IFS-FCC	Proposed
SF	30.4593	73.5011	43.1082	51.4873	70.3214	**78.0013**
EN	0.6578	2.0872	1.7211	2.8742	3.9992	**4.8282**
UIQI	0.2742	0.5577	0.4385	0.7221	0.8314	**0.8934**
AG	8.0973	13.7713	10.1184	15.7832	19.7838	**21.6308**
MI	0.6537	2.1176	1.8722	3.4877	4.8397	**5.2187**
TIME	9.7825	5.8972	13.4862	32.4731	8.1142	**5.0073**

from these modalities plays a pivotal role in distinguishing between active inflammation and chronic demyelination, which is crucial for accurate diagnosis and treatment planning in MS patients. Furthermore, MS lesions can sometimes resemble lesions caused by other neurological conditions or even tumors; thus, fusion aids in distinguishing MS lesions from other types of lesions or tumors, thereby facilitating correct diagnosis and appropriate management. Upon scrutinizing the fusion results of various methods, several vital observations come to light. Firstly, the DWT-IFS fusion method fails to preserve local features, indicating a limitation in accurately retaining fine-scale details from the source images. Following this, the IIFS technique lacks conservation of contour details, implying that the fused images do not accurately represent the boundaries and shapes of objects present in the original images. Moreover, the output image produced by the NSCT-FE method appears blurry, signifying a loss of clarity and sharpness in the fused image, potentially compromising its interpretability and diagnostic utility. Subsequently, the G-CNN-FNN method fails to transfer all the information from the source images to the fused outcome image, indicating a loss of relevant data during the fusion process. However, the IFS-FCC algorithm produces a better-fused result than other methods, but it still needs to improve in preserving minuscule details, suggesting that some finer features are obscured in the fused image. In contrast, the proposed NIFMBF fusion method is highlighted for its ability to incorporate all complementary information from the source images while preserving important features. This indicates that the proposed method effectively combines the strengths of the individual modalities, resulting in a fused image that retains global and local details.

The exact modalities considered in Database-III are also employed for Database-IV, focusing on diagnosing Glioma. This investigation involves analysing images within Pair-8, Pair-9, and Pair-10. T1-weighted MRI is known for its ability to depict contrast enhancement, particularly highlighting areas of the active tumor within the brain. Conversely,

Table 3
The values of fusion metrics for Database-III and Database-IV.

Fusion results for Pair-6						
Metrics	Methods					
	DWT-IFS	IIFS	NSCT-FE	G-CNN-FNN	IFS-FCC	Proposed
SF	8.5247	54.0834	15.1373	20.7821	37.8213	**60.7215**
EN	0.3104	2.7783	1.9345	3.0028	3.5689	**4.6214**
UIQI	0.0942	0.3541	0.1582	0.5217	0.7014	**0.7982**
AG	2.4394	4.8661	3.5324	6.8834	7.4161	**9.9367**
MI	0.3935	2.0176	1.5673	3.2471	4.7466	**5.1078**
TIME	10.2822	6.3872	14.6531	33.4326	9.0078	**5.9523**
Fusion results for Pair-7						
Metrics	Methods					
	DWT-IFS	IIFS	NSCT-FE	G-CNN-FNN	IFS-FCC	Proposed
SF	10.4380	58.8623	17.7782	23.6381	40.5432	**63.9659**
EN	0.2873	2.8341	1.9982	3.1476	3.6400	**4.9681**
UIQI	0.0973	0.3457	0.1973	0.4778	0.6397	**0.7988**
AG	2.9191	4.9789	3.7813	6.9533	8.7102	**9.9932**
MI	0.3514	2.1178	1.6733	3.4827	4.5515	**5.3848**
TIME	11.8214	7.9658	15.8227	35.7443	10.1243	**6.4379**
Fusion results for Pair-8						
Metrics	Methods					
	DWT-IFS	IIFS	NSCT-FE	G-CNN-FNN	IFS-FCC	Proposed
SF	3.2903	36.8846	10.6673	24.2378	35.3773	**38.4172**
EN	0.1867	2.2483	1.0267	2.9732	3.4224	**3.9227**
UIQI	0.0782	0.3434	0.1641	0.4893	0.6972	**0.7243**
AG	0.6617	2.3473	1.4879	3.2936	4.4654	**5.2631**
MI	0.1372	0.4281	0.9783	1.0028	2.9432	**3.4342**
TIME	8.7782	4.2479	12.8942	30.9432	7.3872	**3.9197**
Fusion results for Pair-9						
Metrics	Methods					
	DWT-IFS	IIFS	NSCT-FE	G-CNN-FNN	IFS-FCC	Proposed
SF	3.9689	32.9704	11.0078	25.8273	36.5875	**40.6247**
EN	0.1995	2.1574	1.4504	2.9973	3.5172	**4.0332**
UIQI	0.0836	0.2377	0.1579	0.4891	0.7015	**0.7872**
AG	0.8408	2.3842	1.9267	2.9872	3.8523	**4.3478**
MI	0.4927	2.1439	1.6684	2.9553	3.5251	**4.2873**
TIME	8.8463	4.3842	11.4386	30.5694	7.9378	**3.8211**
Fusion results for Pair-10						
Metrics	Methods					
	DWT-IFS	IIFS	NSCT-FE	G-CNN-FNN	IFS-FCC	Proposed
SF	3.3076	23.0692	9.5509	19.7882	22.1082	**24.8934**
EN	0.7147	1.9673	1.2007	2.5823	3.3608	**4.1358**
UIQI	0.0473	0.2348	0.1035	0.4667	0.6332	**0.7324**
AG	0.7608	4.7827	2.8662	6.3349	11.6914	**12.5432**
MI	0.3773	1.9975	1.5821	2.5724	3.1856	**4.0378**
TIME	8.0847	4.7043	12.9432	31.3476	7.8541	**4.0247**

T2-weighted MRI provides valuable information regarding the extent of edema surrounding the glioma tumor. By integrating information from these modalities, clinicians gain a more comprehensive understanding of the glioma tumor. The fusion of T1-weighted and T2-weighted MRI images facilitates the visualization of both contrast enhancement and morphological changes associated with the tumor. This combined imaging approach enables a more precise assessment of tumor characteristics, including size, shape, and aggressiveness. The fusion results for Database-IV are presented in the last three rows of Fig. 3. The following observations are made based on these results. The fused image produced by the DWT-IFS fusion method exhibits subpar performance. This means that the fusion process leads to a loss of clarity or detail in the final image. In the IIFS method, the edema region associated with the glioma tumor is not correctly identified in the fused image. This limitation affects the ability to assess the extent of edema surrounding the tumor, an important factor in determining disease severity and treatment planning. Next, the output images from the NSCT-FE method display undesirable artifacts. These artifacts compromise the quality and reliability of the fused images. In the G-CNN-FNN method, all image regions, including tumor parts, appear poorly illuminated. This indicates a potential issue with brightness or contrast in the fused image, which impedes accurate visualization of the tumor and surrounding tissues. Further, the textural changes in the original images are not evident in the outcome produced by the IFS-FCC technique. This suggests that the fusion process is ineffective for characterising different tissue types and identifying abnormalities. Nonetheless, the proposed NIFMBF fusion method accurately portrays the correct shape and size of the tumor and edema region with good contrast. This indicates that the proposed method effectively combines information from the T1-weighted and T2-weighted MRI images to produce fused images that maintain important anatomical and pathological features, making them well-suited for clinical analysis and interpretation.

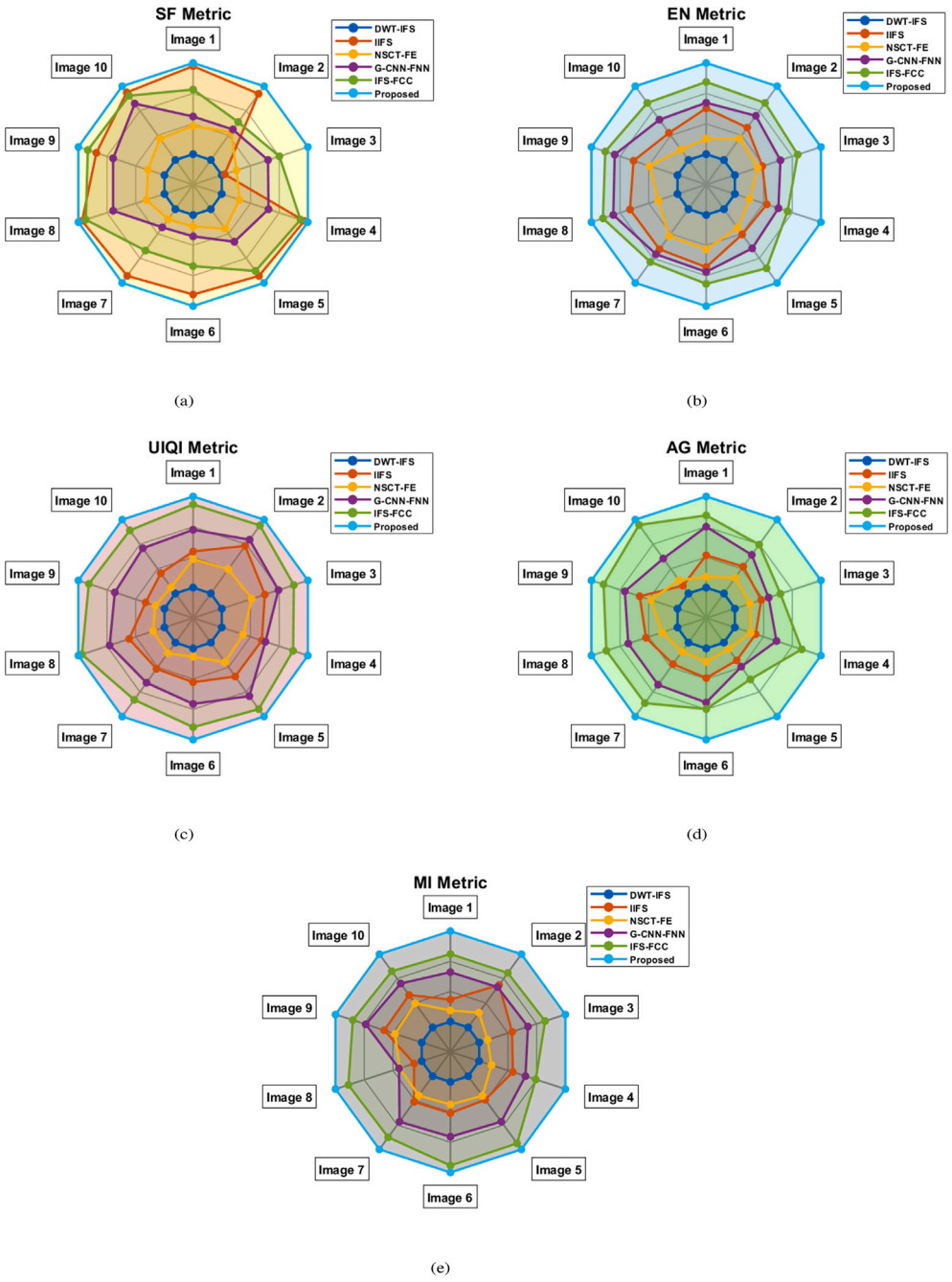


Fig. 4. Graphical representation of fusion metrics for all images.

4.2.2. Quantitative examination

The quantitative analysis is also essential in analysing the results of the proposed fusion method and other comparative methods. While qualitative examination can be subject to human perception and knowledge biases, quantitative investigation offers objective insights based on predefined metrics and considerations of computational complexity. These metrics, elucidated in a prior section, serve as benchmarks for evaluating the performance of fusion methods. The obtained values for each pair, including the proposed method and others, are systematically organised and presented in Tables 2 and 3, facilitating a clear comparison and interpretation of their respective performances. The optimal value for each metric is emphasized in bold to facilitate easier identification and comparison across images. Additionally, the metrics are thoroughly illustrated through graphical representation (refer Fig. 4). The graph reveals that the proposed method outperforms the other metrics, and there are significant variations in the values for the metrics of the other methods. Moreover, Fig. 5 shows a graphical depiction of the average execution time for each algorithm. It is observed that the G-CNN-FNN approach is the slowest due to the combined cost of a Gabor filter bank and deep CNN inference, while NSCT-FE exhibits higher runtimes because the non-subsampled contourlet transform is inherently redundant. Methods based on separable transforms or block-wise IFS operations (DWT-IFS, IIFS, IFS-FCC) show moderate runtimes. Further, the computational complexity of the proposed NIFMBF method is analysed. Let $E = K \times J$ denote the total number of pixels in the image, $n = f \times h$ the block size used for block-wise GLCM computation, and L the number of gray levels. In the Nakagami imaging stage, computing local means or variances and applying the Nakagami transform requires

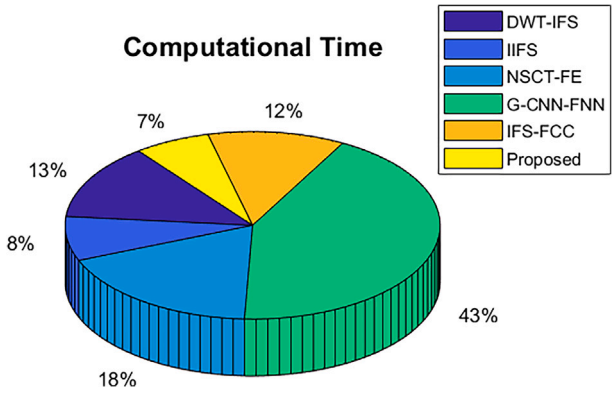


Fig. 5. Graphical representation of average computational time for every fusion methods.

a constant number of operations per pixel, resulting in a complexity of $\mathcal{O}(E)$. Similarly, the channel-wise IFI transformation also operates linearly on the pixels, giving $\mathcal{O}(E)$. For GLCM-based contrast extraction, approximately $\frac{E}{n}$ blocks are processed. Each block requires computing a GLCM of size $L \times L$ and extracting contrast, which incurs a cost of $\mathcal{O}(L^2 n)$ per block. Thus, the total cost becomes $\mathcal{O}(\frac{E}{n})\mathcal{O}(L^2 n) = \mathcal{O}(EL^2)$. Finally, the block fusion and reconstruction steps involve a constant number of operations per pixel, contributing an additional $\mathcal{O}(E)$. Combining all stages, the overall computational complexity of the proposed method is $\mathcal{O}(EL^2) + \mathcal{O}(E) = \mathcal{O}(EL^2)$. Although the GLCM-based contrast extraction in the proposed NIFMBF can be expensive in theory, the proposed method produces the lowest average runtime compared to other methods. Further, a detailed examination of various fusion techniques based on the graphical representations is summarized below.

(iii) DWT-IFS: Performs poorly across all metrics compared to other methods. Although its execution time is relatively short, its overall fusion performance is weak.
(iii) IIFS: Produces low metric values, primarily due to over-enhancement in the images. While computational time is minimal, the quality remains unsatisfactory.
(iii) NSCT-FE: Falls below expectations in performance and suffers from very high computational cost, which further degrades its applicability.
(iv) G-CNN-FNN: Achieves average metric values; however, the algorithm's running time is considerably high, making it less practical for real-time applications.
(v) IFS-FCC: Delivers better fusion metric outcomes than several alternatives, but its computational time is slightly longer.
(vi) Proposed NIFMBF: Demonstrates superior performance across all metrics for every image while retaining significantly reduced computational time compared to all other fusion methods.

5. Discussion

From the above observations, the qualitative examination suggests that the NIFMBF fusion method produces visually appealing results. This method likely preserves essential details and features from the input images, resulting in fused images that are clear and informative. Quantitative analysis using predefined metrics demonstrates that the NIFMBF fusion method achieves superior performance across these metrics. This indicates that the fused images generated by this method exhibit good results compared to alternative fusion techniques. Additionally, the NIFMBF fusion method stands out in terms of computational efficiency. This implies that the method is effective in producing high-quality fused images and achieves this result promptly. Thus, the combined qualitative and quantitative examination highlights the NIFMBF fusion method as an auspicious approach for brain image fusion tasks. To evaluate the efficiency of the proposed method, a brief sensitivity analysis was conducted on the key parameters used in the framework. The proposed NIFMBF framework is influenced primarily by two parameters: the Nakagami parameter used in the NI stage and the block size used for GLCM computation. A sensitivity assessment was performed by varying these two parameters within practical ranges. The results indicate that NIFMBF maintains stable performance across moderate variations of $\pm 10 - 15\%$ in these parameters, with changes in fusion metric values remaining within a $2 - 5\%$ deviation range. The resulting variations in fusion quality metrics remained small, indicating that the method is robust to moderate parameter changes. Moreover, entropy values are optimized internally during IFI computation. This optimization is automatic and does not require external tuning. Further, to evaluate the contribution of each component in the proposed NIFMBF framework, an ablation study was conducted under three configurations: (i) without NI, (ii) without the proposed IFG, and (iii) without GLCM-based contrast features. Pair-6 was considered for this study to analyse the importance of each component and the variations in the results. The visual comparison of the output images is shown in Fig. 6. It can be observed that the fused image without NI fails to highlight subtle regions, the output without IFG appears less balanced with uneven contrast, and the fusion without GLCM features produces blurred edges with a loss of fine textural details. The quantitative results obtained for this ablation study are summarized in Table 4. It confirms that removing any single component reduces the overall improvement in image quality. In contrast, the output of the proposed NIFMBF method discussed already successfully preserves structure, enhances contrast, and retains critical details across modalities. Furthermore, to evaluate the robustness of the proposed NIFMBF framework under noisy conditions, 5% Gaussian noise was artificially added to the input images of Pair-3. The noisy source images and the corresponding fused result are shown in Fig. 7, while the quantitative metric values are provided in Table 5. Performance metrics show a slight decline under noise inter-

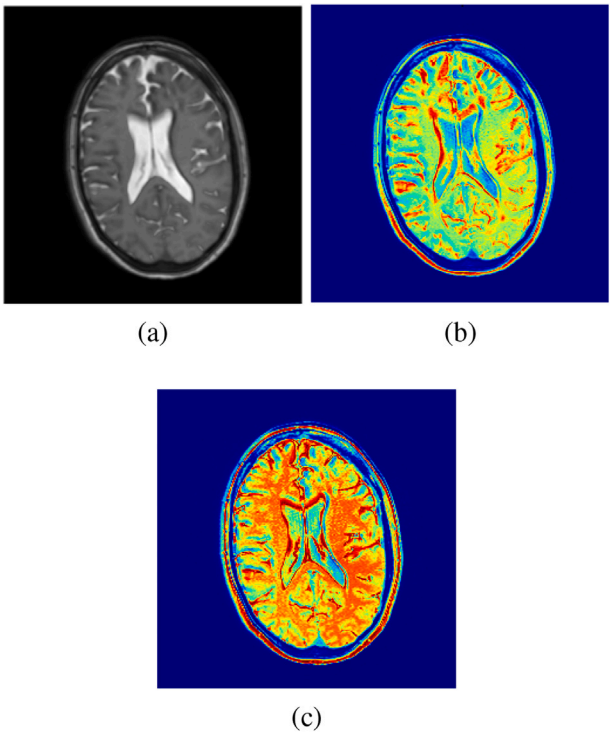


Fig. 6. Visual results of the ablation study: (a) Fused image without NI, (b) fused image without the proposed IFG, (c) fused image without GLCM-based contrast features.

Table 4
Quantitative results of the ablation study.

Metrics	Variants		
	Without NI	Without IFG	Without GLCM
SF	36.9247	40.0234	49.1563
EN	3.2546	2.5783	3.5345
UIQI	0.6425	0.5545	0.6152
AG	6.8494	6.1651	5.8326
MI	4.9135	4.3172	3.9673

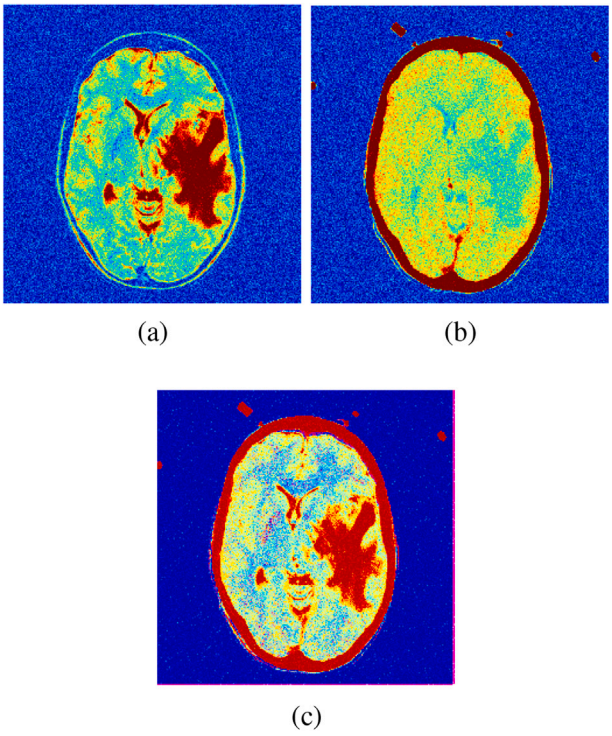


Fig. 7. Visual results of Pair-3 under 5% Gaussian noise: (a) CT image, (b) MRI image, and (c) fused output obtained using the proposed method.

Table 5
Quantitative results of Pair-3 under 5% Gaussian noise for the proposed method.

Input	Metrics				
	SF	EN	UIQI	AG	MI
Pair-3	63.5242	4.0254	0.8567	11.5731	4.9563

ference. However, the NIFMBF consistently preserves structural details and information content. Thus, these results, together with the ablation study, further demonstrate the efficiency and reliability of the proposed method. Moreover, clinical implications are essential in medical image fusion research because they connect the algorithmic advancements with real-world use and enhance the significance of the study. The proposed NIFMBF method yields improved contrast, enhanced lesion visibility, and clearer delineation of anatomical structures. All of these features have significant clinical implications. By providing a fused image that simultaneously highlights pathological regions and preserves normal structural boundaries, the method can support radiologists in making confident diagnostic interpretations. The clarity of fused information from modalities such as T1-weighted MRI and T2-weighted MRI can also aid in pre-surgical assessment by providing better localization of abnormal tissue, improved boundary definition, and enhanced visualization of critical regions. Thus, the enhanced fusion output has the potential to improve both diagnostic accuracy and surgical planning workflows.

6. Conclusion

The NIFMBF fusion method was developed to overcome existing challenges in brain image fusion, particularly concerning uncertainty in low-illumination scenarios. To achieve these improvements, statistical enhancement using NI was employed to highlight low-contrast and suspicious regions in brain images to improve the visibility of subtle structures prior to fusion. A new IFG was developed specifically for image fusion that effectively reduces uncertainty and enhances spatial information. Furthermore, contrast features extracted via GLCM were used to guide fusion, ensuring that fine details were preserved while redundancy was minimised. Thus, the proposed NIFMBF method enhances the clarity and detail of fused brain images. Moreover, the efficacy of the proposed method was evaluated across four diverse databases, demonstrating its versatility and applicability across different types and dimensions of brain image data. To gauge its practicality, the proposed method was compared against five benchmarking fusion techniques, providing a robust analysis of its performance. Both qualitative and quantitative examinations were conducted for all fusion methods, ensuring a comprehensive assessment. The experimental results confirm that the NIFMBF delivers a consistent quantitative average gain of 12–20% for every fusion metric over five state-of-the-art fusion techniques. Furthermore, it achieves a 30% reduction in computation time. These improvements highlight the robustness, effectiveness, and computational advantage of the proposed framework. Moreover, the sensitivity analysis, ablation study and the noise analysis further demonstrate the reliability of the proposed method. Furthermore, the IFG in this study is developed specifically for the fusion of medical images, ensuring accurate and reliable fusion outcomes. However, the absence of an integrated segmentation component limits its applicability to region-based clinical analysis. Accordingly, future work is directed towards developing segmentation processes that combine Nakagami imaging with IFS, offering promising advancements in brain imaging and incorporating medical-specific metrics, such as lesion detectability, to enhance clinical relevance.

CRediT authorship contribution statement

K.G. Lavanya: Writing – original draft, Validation, Software, Methodology, Investigation. **P. Dhanalakshmi:** Supervision, Methodology, Investigation, Conceptualization. **M. Nandhini:** Writing – original draft, Software, Methodology, Investigation.

Ethical approval

This article does not contain any studies involving human participants conducted by any of the authors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Enquiries about data availability should be directed to the authors.

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